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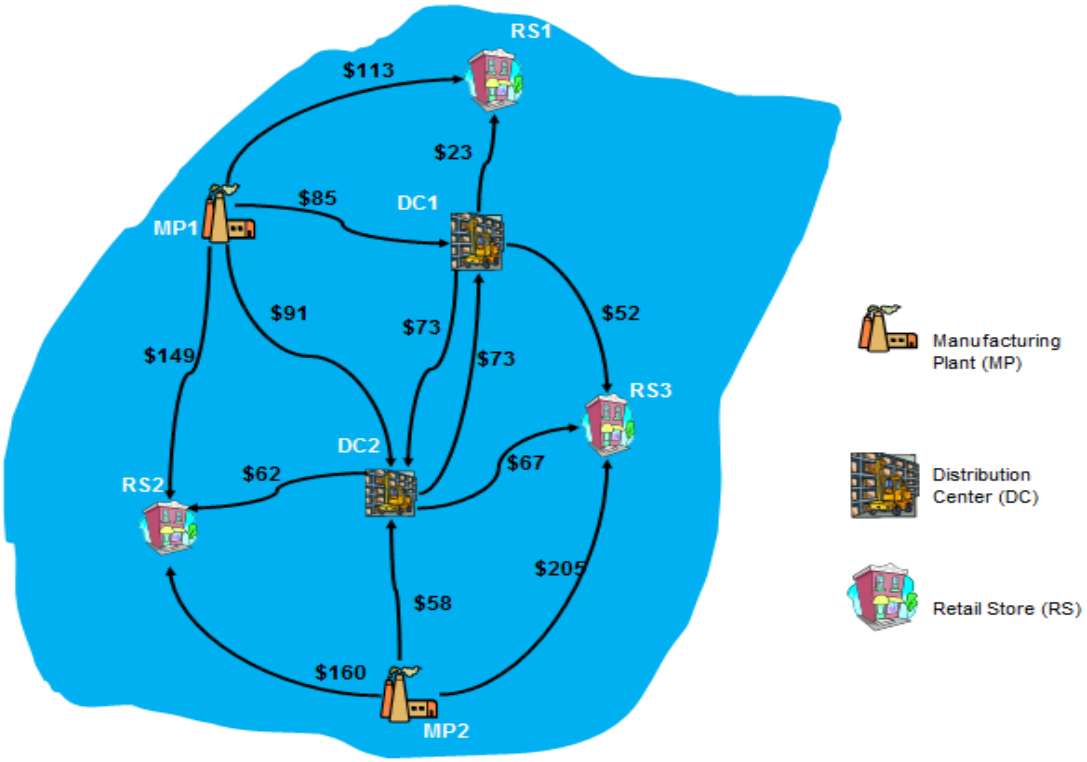
AY 2023 E211 Operations Planning II
ESE Practice Questions

Question 1

D&T has two manufacturing plants MP1 and MP2 producing a newly developed gadget. The product is shipped to three retailers RS1, RS2 and RS3 directly or through two distribution centers DC1 and DC2. The distribution network available for shipping the product is illustrated in the following figure where the numbers on the arcs represent the unit shipping cost.



The supplies from the two plants are 400 (MP1) and 630 (MP2) units, and the demands from the three retailers are 360 (RS1), 250 (RS2) and 260 (RS3) units for this month. The distribution center 2 (DC2) also has a demand of 160 units. In addition, the shipment cannot exceed 180 units from MP2 to DC2, and 130 units from MP1 to DC1. Help D&T develop a Linear Programming (LP) model to determine an optimal distribution plan.

Shown below is the network representation of the distribution problem.

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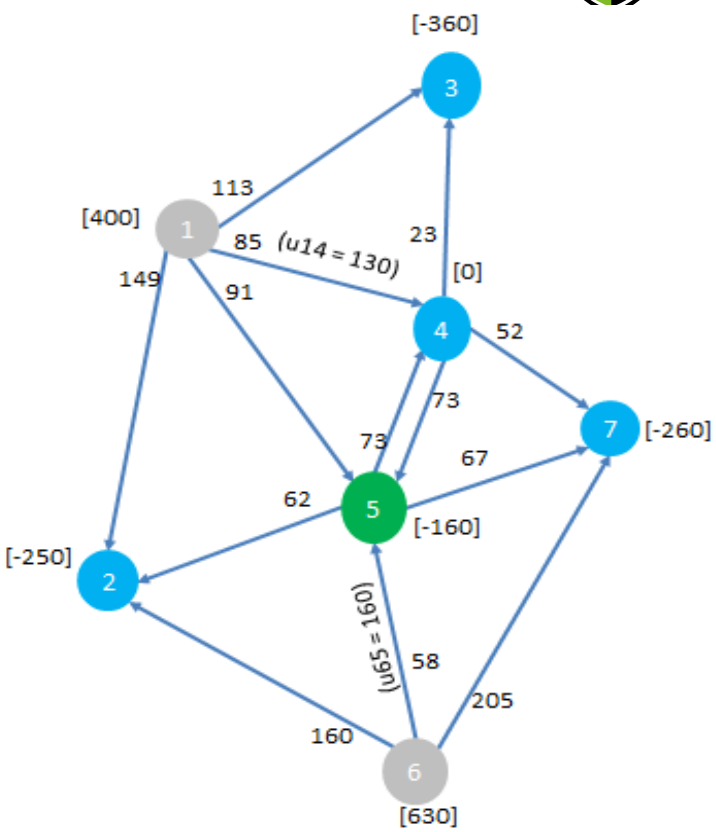


a) Define the decision variables.

Let Xij represents the number of products to be shipped from node I to node i.
I, I = 1,2,3,4,…..7

b) What is the objective of the LP model? Write down the objective function.

Objective is to minimize the total cost of shipping products from the manufacturing plants to the demand centers.

Min. Cost = 149X12 + 113X13 + 85X14 + 91X15 + 160X62 + 73X54 + 73X54 + 23X43 + 52X47 + 67X57 + 160X65 + 205X76

c) Formulate the constraints.

Amount shipped out – Amount shipped in = Net flow at the node

Node 1: X12 + X13 + X14 + X15 = 400
Node 2: -X12 – X52 – X62 = -250
Node 3 = -X13 – X43 = -360
Node 4 = X43 + X45 + X47 – X14 – X54 = 0
Node 5 = X57 + X52 + X54 – X14 – X54 = 0

Node 6 = X62 + X65 + X67 = 630
Node 7 = -X47 – X57 – X67 = -260

*Singe total supply (1030) = total demand (1030), the '=' sign is used in the flow constraint equations.

If total supply (1100) > total demand, use '<=/ sign for all supply nodes.
E.g., Node 1: X12 + X13 + X14 + X15 <= 400

If total demand > total supply, use '>=' sign for all demand nodes.
E.g. Node7: $-X_{47} - X_{57} - X_{67} >= -260$

Upper bound constraints

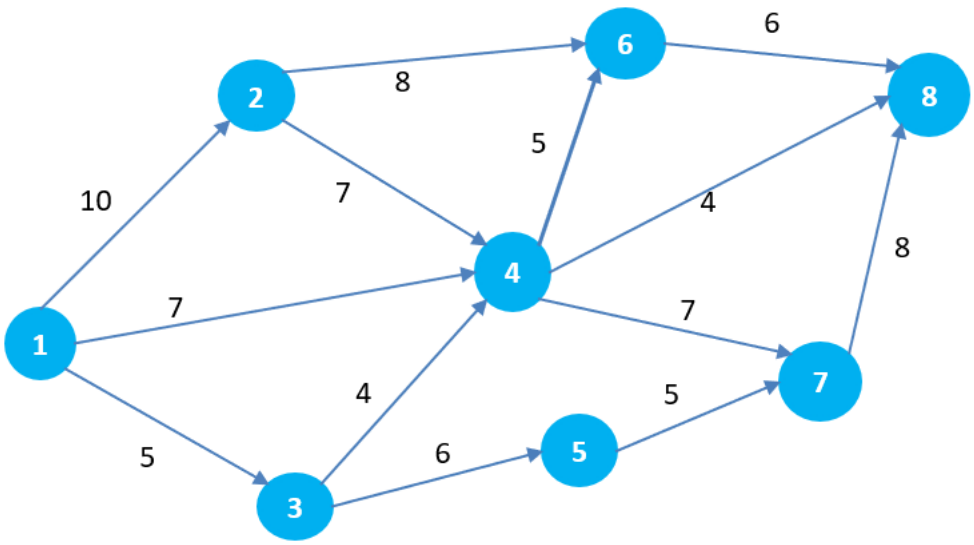
Flow through an arc cannot exceed the arc capacity

$X_{14} <= 130$
$X_{65} <= 180$

Non-negativity constraints
All $X_{ij} >= 0$

Question 2

Determine the shortest path from node 1 to each of the other nodes using Dijkstra's algorithm. The values on the arcs represent the distance between 2 nodes.



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Iteration	Solution Set	Arcs	Total Distance
Iteration 1	1	1 → 2 1 → 3 1 → 4	10 5 7
Iteration 2	1 3 (5)	1 → 2 1 → 4 3 → 4 3 → 5	10 7 9 11
Iteration 3	1 3 (5) 4 (7)	1 → 2 3 → 5 4 → 6 4 → 7 4 → 8	10 11 12 14 11
Iteration 4	1 3 (5) 4 (7) 2 (10)	3 → 5 4 → 6 4 → 7 4 → 8 2 → 6	11 12 14 11 18
Iteration 5	1 3 (5) 4 (7) 2 (10) 5 (11)	4 → 6 4 → 7 4 → 8 2 → 6 5 → 7	12 14 11 18 16
Iteration 6	1 3 (5) 4 (7) 2 (10) 5 (11) 8 (11)	4 → 6 4 → 7 2 → 6 5 → 7	12 14 18 16
Iteration 7	1 3 (5) 4 (7) 2 (10) 5 (11) 8 (11) 6 (12)	4 → 7 5 → 7	14 16
Iteration 8	1 3 (5) 4 (7) 2 (10) 5 (11) 8 (11) 6 (12)	7 (14)	

4 + 5

Question 3

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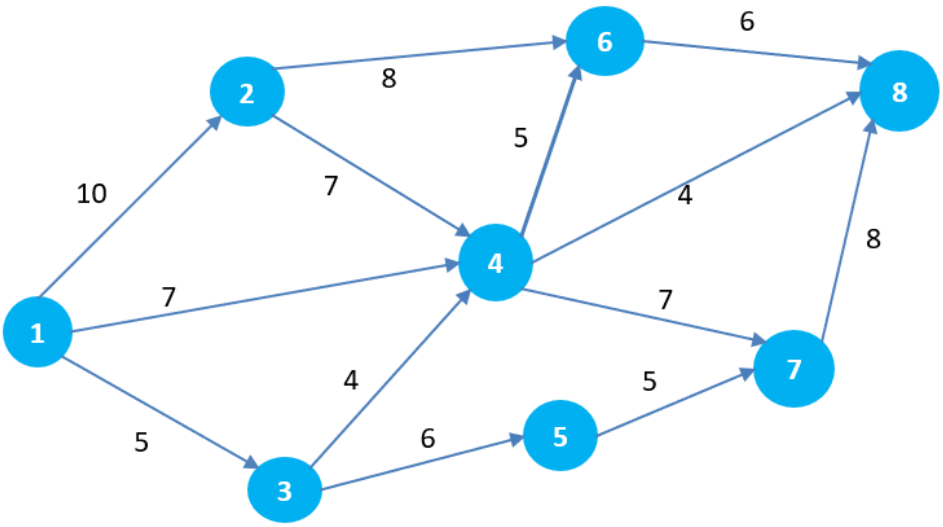
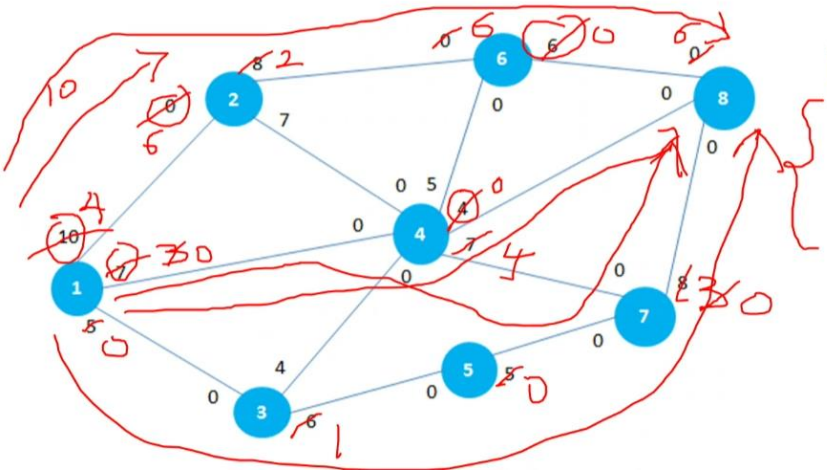
Determine the maximum flow (Ford-Fulkerson's) from source node 1 to destination node 8. Show your step-by-step workings clearly. The values on the arcs represent the capacity between 2 nodes along the direction of flow.



*Always pick the smallest number in a path/ arcs

Q3 Ford Fulkerson's algorithm



Path	Flow
1-2-6-8	6
1-4-8	4
1-3-5-7-8	5
1-4-7-8	3

Max flow = 18

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Question 4

On a certain day, you need to formulate a plan to deliver a certain product (packed in boxes) from warehouse (1) to six customers (2, 3, 4, 5, 6, 7). The following table shows the distance (in km) between the warehouse and each customer.

	1	2	3	4	5	6	7
1	-	26	14	30	28	29	31
2	26	-	15	23	25	8	13
3	14	15	-	14	16	15	11
4	30	23	14	-	18	21	35
5	28	25	16	18	-	33	23
6	29	8	15	21	33	-	9
7	31	13	11	35	23	9	-

a) Is this a routing or a scheduling problem? Explain.

Routing. It involves forming the sequence in which locations are to be planned to deliver a certain product.

b) Apply the Nearest Neighbour method to develop a delivery route.

1 – 3 -7 – 6 - 2 - 4 - 5 - 1

c) Apply the Clarke & Wright Savings method to plan a delivery route.

Sij	1	2	3	4	5	6	7
1	0	0	0	0	0	0	0
2	0	0	25	33	29	47	44
3	0	25	0	30	26	28	34
4	0	33	30	0	40	38	26
5	0	29	26	40	0	24	36
6	0	47	28	38	24	0	51
7	0	44	34	26	36	51	0

i. Verify the Savings value for the 3-4 pair and 2-7 pair.

3 -> 4 = C13 + C14 – C34 = 14 + 30 – 14 = 30
2 -> 7 = C12 + C17 – C27 = 26 + 31 – 13 = 44

ii. Using the ranked Savings table given, construct the delivery route, showing clearly the sub-tour(s) formed at each step.

Rank	Node Pair	Savings
1	6--7	51
2	2--6	47
3	2--7	44
4	4--5	40
5	4--6	38
6	5--7	36
7	3--7	34
8	2--4	33
9	3--4	30
10	2--5	29
11	3--6	28
12	3--5	26
13	4--7	26
14	2--3	25
15	5--6	24

*Cannot break the bond between existing pair

Rank	Path (Tour)	Remark
1	1—6-7-1	
2	1-2-6-7-1	
4	1-2-6-7-1 , 1-4-5-1	
6	1-2-6-7-5-4-1	
9	1-2-6-7-5-4-3-1	

d) Compare the total travel distance for the routes derived from the Nearest Neighbour method and Clarke & Wright Savings method.

Question 5

As an operations manager, Mr Lee needs to plan the amount of a certain product to produce according to a monthly forecast derived from the historical sales. Mr Lee has collected the following data for the past 12 months:

Month	Sales for the product (kg)
1	1025
2	1250
3	1115
4	955
5	1200
6	1350
7	1025
8	1250
9	1150
10	1350
11	1050
12	1150

Show your workings clearly and round off your answers to **two** decimal places for the following questions.

a) Calculate the forecast for month 13 based on three-month and five-month moving average. Do you choose a three-month moving average or a five-month moving average if you want your forecast to be **less responsive** to the most recent data?

F_{13} (Three-month moving average) = (1150 + 1050 + 1350)/ 3
= 1183.33

F_{13} (Five-month moving average) = (1150 + 1050 + 1350 + 1150 + 1250)/ 5
= 1190.00

Choose the **five-month moving average** (higher N) forecast for the forecast to be less responsive to the most recent data.

*If you want forecast to be less responsive = smoother, hence take the larger period.

b) Given the weights of 8, 5 and 2, calculate the three-month weighted moving average forecast for month 13.

F_{13} (Three-month moving average) = (1150 + 1050 + 1350)/ 3
= 1183.33

F_{13} (Five-month moving average) = (1150 + 1050 + 1350 + 1150 + 1250)/ 5
= 1190.00

Choose the **five-month moving average** (higher N) forecast for the forecast to be less responsive to the most recent data.

F_{13} (Three-month weighted moving average)
= (1150 x 8 + 1050 x 5 + 1350 x 2)/(8+5+2)
= 1143.33

c) Given the simple exponential smoothing forecast for months 9 through 12, using Alpha (α) of 0.2 and 0.4, compute:

Month	Sales for the product (kg)	Exponential Smoothing (Alpha = 0.2)	Exponential Smoothing (Alpha = 0.4)
9	1150	1141.48	1180.29
10	1350	1143.18	1168.17
11	1050	1184.55	1240.90
12	1150	1157.64	1164.54
13		????	?????

i. Forecast for month 13 using simple exponential smoothing with alpha (α) of 0.2

F_{13} (Simple exponential smoothing α=0.2)
= 1150 x 0.2 + 1157.64 x (1-0.2) = 1156.11

ii. The forecast for month 13 using simple exponential smoothing with alpha (α) of 0.4

F_{13} (Simple exponential smoothing α=0.4)
= 1150 x 0.4 + 1164.54 x (1-0.4) = 1158.72

iii. Which forecast is less responsive to the most recent data? Explain.

0.2 SES. The smallest forecast won't be affected since the value is rather small.

d) Mr Lee is told that for the next 12 months, the sales of the product are likely to decrease. Should any of the methods above (Moving Average, Weighted Moving Average or Simple Exponential Smoothing) be used to forecast the sales?

None, lag.

e) Calculate the Absolute Percent Error for Month 11 in the table given.

Month	Sales for the product (kg)	Forecast by Exponential Smoothing (kg)
7	1025	1136.69
8	1250	1114.35
9	1150	1141.48
10	1350	1143.18
11	1050	1184.55
12	1150	1157.64

Absolute Percent Error = abs(difference between actual and forecast)/(actual) *100%

= |1050-1184.55|/1050 * 100% = 12.81%

8	1250	1114.35
9	1150	1141.48
10	1350	1143.18
11	1050 ✓	1184.55 ✓
12	1150	1157.64

Question 6

Amy is using the multiplicative decomposition method to forecast based on weekly sales data from the months September to November. The following table shows the Excel template she used to do her forecast.

Show your workings clearly and round off your answers to **two** decimal places for the following questions.

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Week	Actual value	Time period	MA	CMA	Seasonal ratio	Seasonal Index	Deseasonalized value
SEPW1	256	1				0.86	299.28
SEPW2	322	2	381.00			0.90	356.92
SEPW3	368	3	415.75	398.38	0.92	0.94	391.15
SEPW4	578	4	A	433.00	1.33	1.30	444.06
OCTW1	395	5	492.00	471.13	0.84	0.86	461.78
OCTW2	460	6	536.25	B	0.89	0.90	509.88
OCTW3	535	7	586.75	561.50	0.95	0.94	568.65
OCTW4	755	8	634.00	610.38	1.24	1.30	580.05
NOVW1	597	9	675.25	654.63	C	0.86	697.93
NOVW2	649	10	721.50	698.38	0.93	0.90	719.37
NOVW3	700	11				0.94	D
NOVW4	940	12				1.30	722.18
						intercept	265.90
						slope	42.37

a. State the number of averaging periods used to calculate the MA.

N = Even = 4
*Look at the weeks

b. Calculate the missing entries illustrated as A, B, C and D in the Excel template.

c. Calculate the seasonalized forecast for Week 2 of December:

i. What is the linear equation used to forecast the deseasonalized values?

Y = 265.9 + 42.37x

ii. Calculate the deseasonalized forecast for Week 2 of December.

Index period for Week 2 of December is 14.

Deseasonalized forecast for Week 2 of December,

Y = 265.9 + 42.37* 14 = 859.08

iii. Calculate the seasonalized forecast for Week 2 of December.

Seasonalized forecast for Week 2 of December,

F_{14} = 859.08 * 0.90 = 773.17

Question 7

GoodHealth is a clinic with one duty doctor. On average, the doctor needs to spend about 5 minutes on each patient. The average number of patients arriving at the clinic every hour is 10.

Show your workings clearly and round off your answers to **two** decimal places for the following questions

a. State the queuing model you can use to calculate the queue performance measures at the clinic. What are the assumptions needed?

M/M/1

b. Calculate the mean arrival rate and mean service rate in per minute.

Patients arrival per minute = 60 /10 = 6

Arrival rate = 1/ λ
= 1/ 6
= 0.1666 ~ 0.167

OR

Arrival rate = 10./60 per minute

Service rate = 1/ c
= 1/ 5
= 0.2

c. Calculate the utilization, ρ.

P = λ/ μ
= 0.167/ 0.2
= 0.83

d. Calculate the average number of patients waiting for the doctor, Lq.

$$L_q = \frac{\lambda^2}{\mu(\mu-\lambda)} = \frac{\rho^2}{1-\rho}$$

0.83 ^ 2 / (1-0.83)ss

= 4.05

e. Calculate the average time a patient spends at the clinic, Ws.

$$W_q = \frac{L_q}{\lambda}$$

4.05 / (1/6) = 24.3

$$W_s = W_q + \frac{1}{\mu}$$

24.3 + 5
= 29.3

1/ μ is service rate

f. Suppose the patient arrival rate increases to 23 per hour during the flu season. What is the minimum number of doctors required for the queueing system to be in a steady state condition? Average service time remains at 5 minutes.

Patients arrival per minute = 60 /23 = 2.608
Arrival rate = 1/ λ
= 1/ 2.608
= 0.383

OR

Arrival rate = 23/ 60
= 0.383

Service rate = 0.2

*In order to reach steady state, effective arrival rate (λ) must be less than sum of effective service rates (μ).

0.2k > 0.383
K > 0.383/ 0.2
K. > 1.915
K = 1.915 ~ 2 (minimum)

g. How can a M/M/k model be used to design the clinic operations?

M/M/K model can be used to design the clinic operations by having multiple duty doctor to attend to the patients.

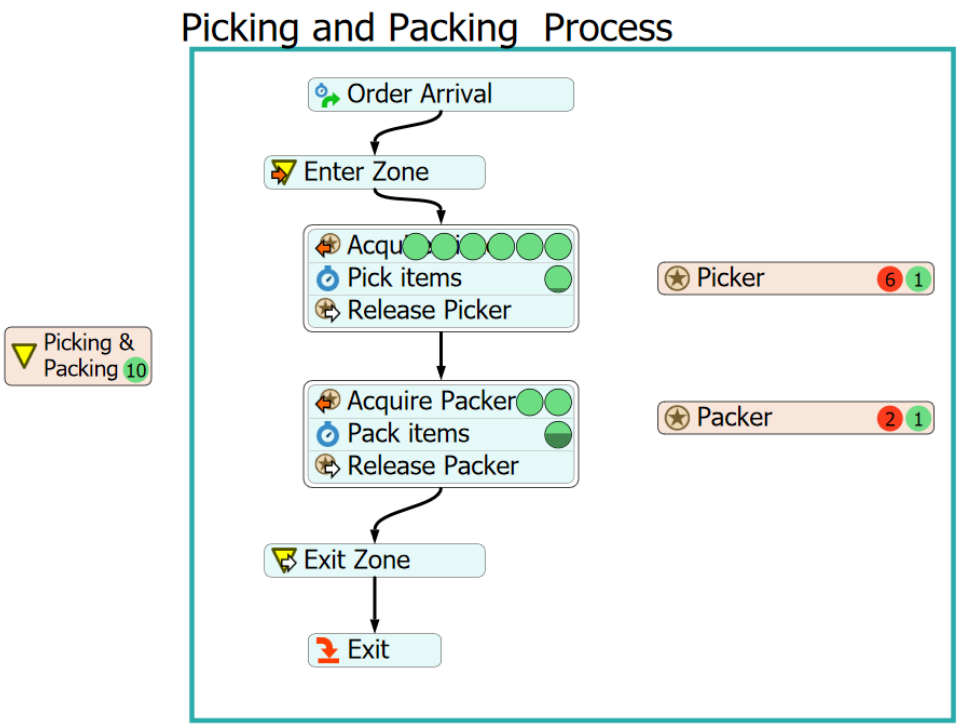
The M/M/k model steady state condition can be used to:
- Determine the number of doctors required
- Determine the service time standard
- Determine the maximum patient arrival rate that can be handled by the clinic.

Question 8

A process flow simulation model was built for a warehouse picking and packing process as shown in the figure below.

a. State the number of pickers in the process and the number of items waiting to be picked.

Number of pickers = 1
Number of items waiting to be picked = 6



b. What is the purpose of the activities 'Acquire Packer' & 'Release Packer'?

Acquire packer: Obtain packing resources before packing can begin.

Release packer: Return the packer resource to the available state after the item has been packed

c. Figure below shows the statistics of the Picker.

i) What is the Average and Max number of orders waiting for the Picker?

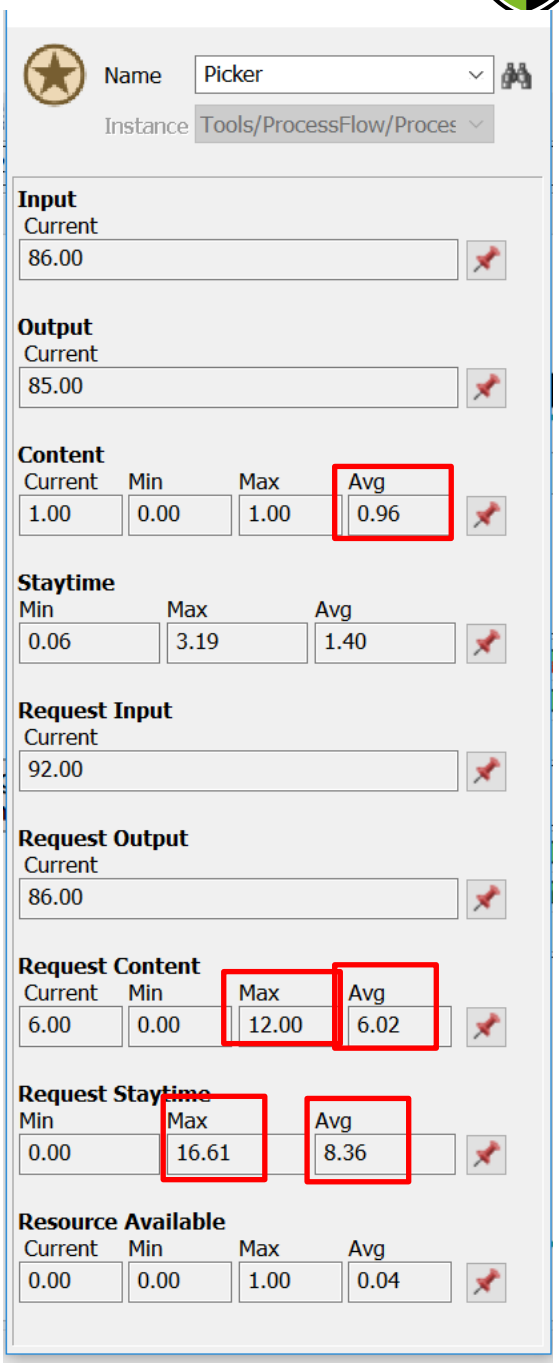
Average = 6.02
Max number of orders = 12.00

ii) What is the Average and Max waiting time for the Picker?

Average = 8.36
Max waiting time = 16.61

iii) What is the average utilization of the Picker?

Average content / max resource available = 0.96 /1 = 0.96



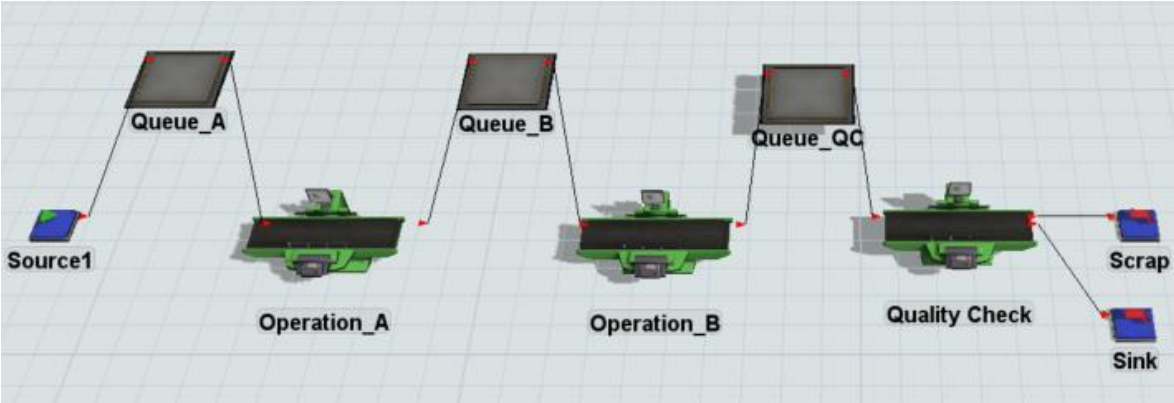
d. How would you know who is the bottleneck, Picker or Packer?

To identify the bottle neck., compare the number in the queue for picker and packer. Such as the queue length and waiting time, utilisation.

Question 9

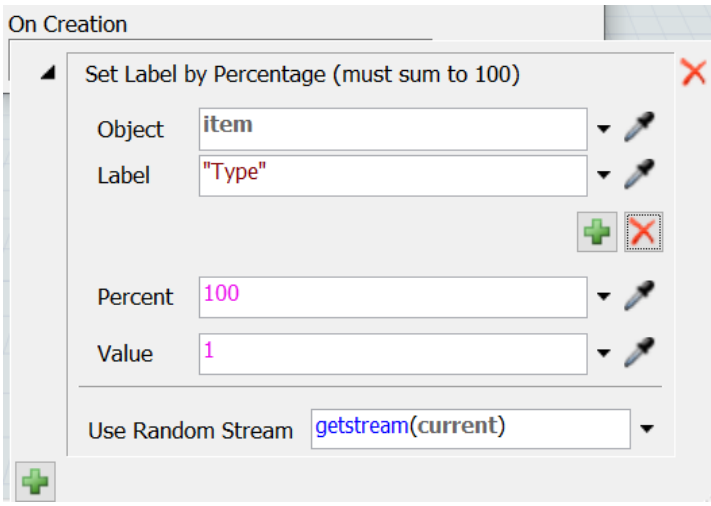
A production line has three major operations: Operation_A, Operation_B, and Quality Check.

a. A simulation model is built as shown. Suppose there is limited space for the work-in-progress parts in the production line. How do you model this?



Go to the queue property and set the max-content. If the space can only hold up to 20 items, then the maximum content should be 20.

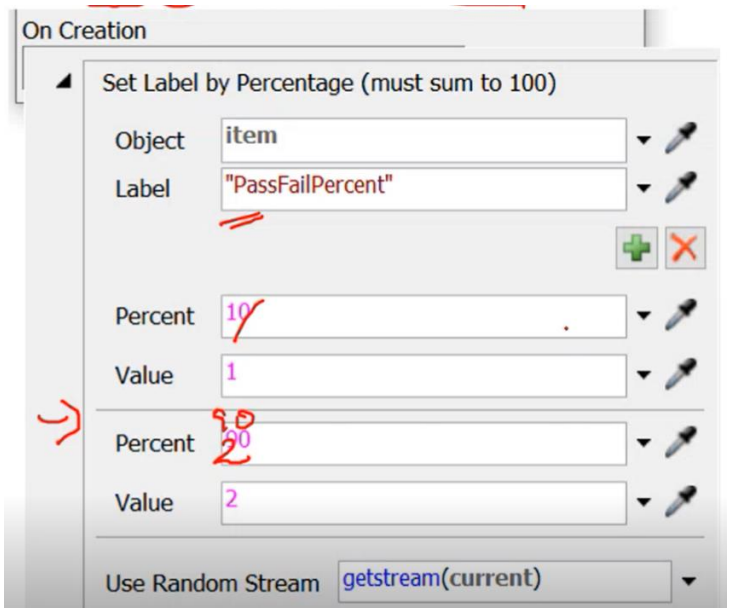
b. After the products go through Quality Check, they are either sent to Scrap (failed QC) or Sink (passed QC). The percentage of products passing QC is 90%. To model this, a label named 'PassFailPercent' is to be created for each item. Items with label value of 1 will be sent to Scrap whereas those with label value of 2 will be sent to Sink. How do you set this at the On Creation trigger of Source1?



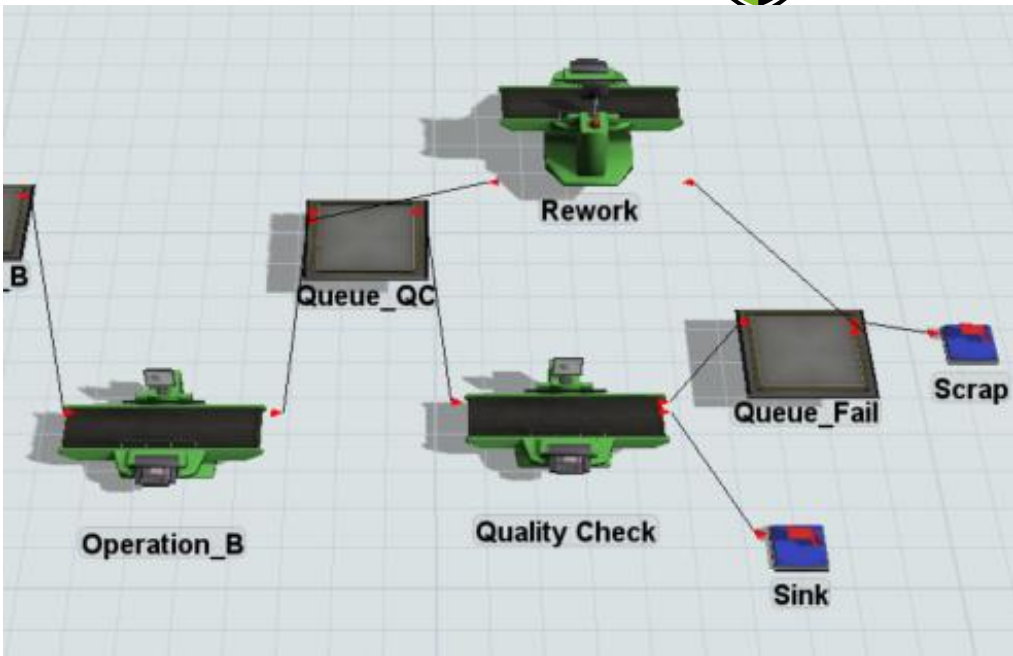
Label: PassPercentFail

Percent: 10
Value: 1

Percent: 90
Value: 2

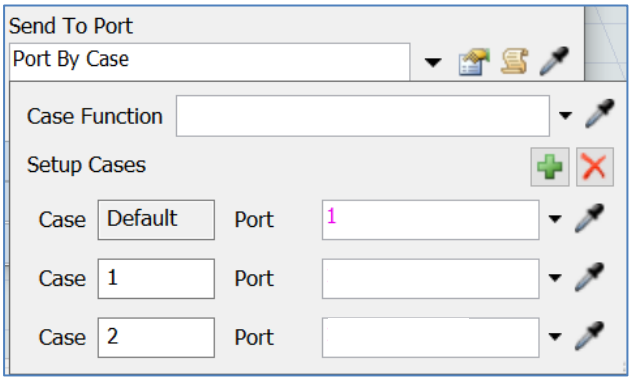


c. To reduce the percentage of products being scrapped, only products that failed the Quality Check twice will be sent to the Scrap; those failed once will be sent to Rework station and then routed back to Quality Check.



An item label 'Qcheck' is created to track the number of times that a product has gone through Quality Check. The label is initialized with a value of zero. Each time the item passes through Quality Check, the label value will be incremented by 1.

At Queue_Fail, 'Port By Case' option is used to direct the flow of the products. Fill in the blanks below.

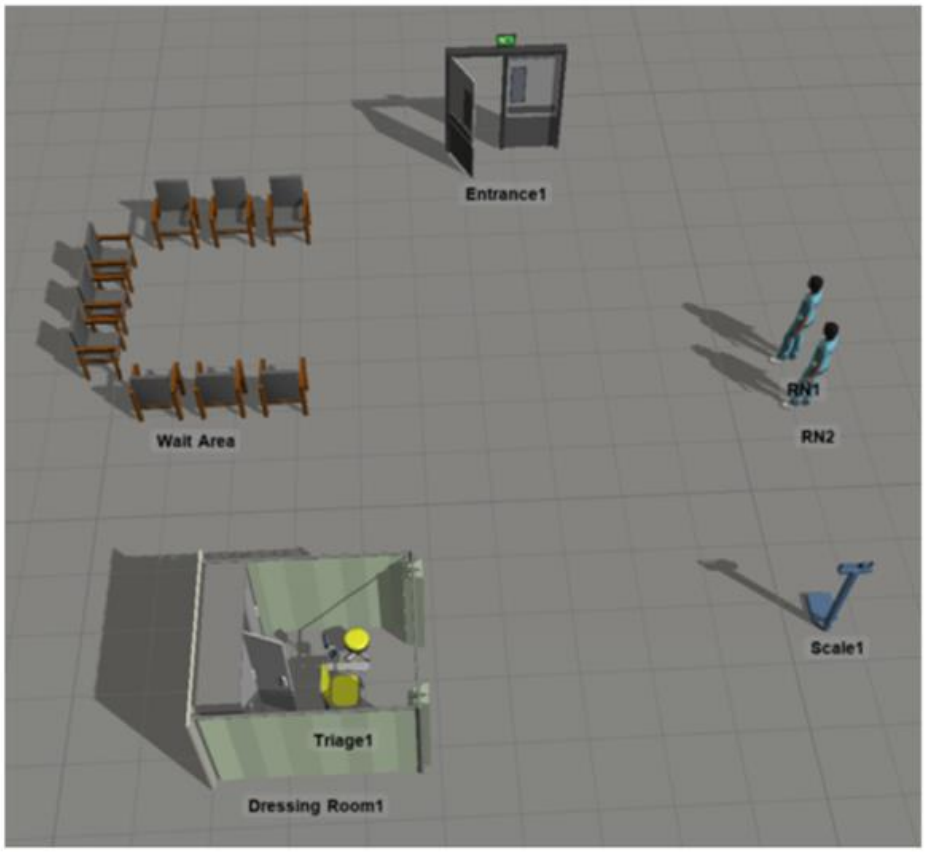


Case Function = item.Qcheck

Case 1, Port = 2

Case 2, Port = 1

Question 10



a) Where in FlexSim 3D Healthcare library do you find:

i. Entrance 1
ii. RN1
iii. Scale1
iv. Triage1

i. Prop
ii. Staff
iii. Location
iv. Location

b) Where do you key in the arrival data of patients?

Under Patient Flow, Arrivals tab, Source Arrival Table

c) Which activity sets do you use to model:

i. Patients walk from Wait Area to Scale and take measurements themselves.

   Walk then process

ii. RN escorts Patient from Wait Area to Triage and check on patient's condition.

   Escort then process

iii. RN walk to patient at waiting area to check on patient's condition.

   Process with staff

iv. Patients walk to Triage. However, if Triage is busy, patients will walk to waiting area to wait.

   Walk or wait area

v. RN takes a wheelchair to move Patient from waiting area to triage before checking on patient's condition

   Transport then process

vi. RN escorts patient out through Entrance1

   Escort to exit